

ID: _____

NAME: _____

QUIZ 7-A

SECTION: _____

Instruction: No marks will be awarded for Direct Answers

1. Process the JK Latch given below for all possible cases and fill the table accordingly?

CIRCUIT:

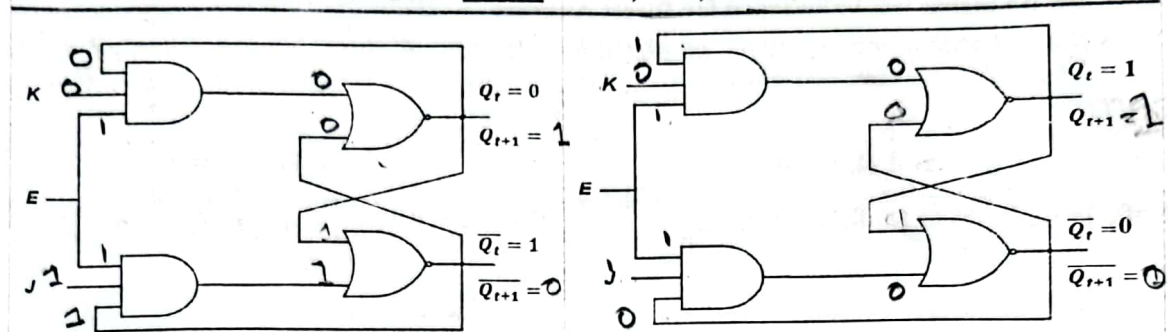
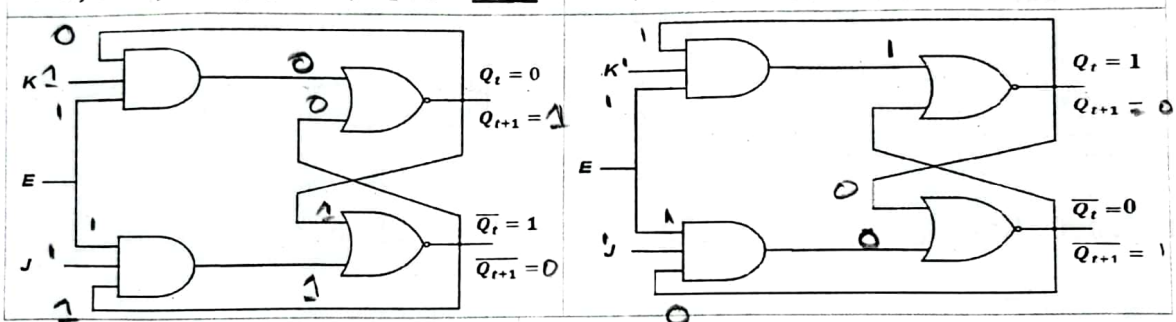
Case-I ($Q_t = 0$)	Case-II ($Q_t = 1$)
$E=0, J=\times, K=\times, Q_t=0, Q_{t+1} = \underline{0}$	$E=0, J=\times, K=\times, Q_t=1, Q_{t+1} = \underline{0}$
$E=1, J=0, K=0, Q_t=0, Q_{t+1} = \underline{0}$	$E=1, J=0, K=0, Q_t=1, Q_{t+1} = \underline{1}$
$E=1, J=0, K=1, Q_t=0, Q_{t+1} = \underline{0}$	$E=1, J=0, K=1, Q_t=1, Q_{t+1} = \underline{0}$

ID:

NAME:

QUIZ 7-A
Saliq loban

SECTION:

 $E=1, J=1, K=0, Q_t=0, Q_{t+1} = \underline{1}$
 $E=1, J=1, K=0, Q_t=1, Q_{t+1} = \underline{1}$

 $E=1, J=1, K=1, Q_t=0, Q_{t+1} = \underline{1}$
 $E=1, J=1, K=1, Q_t=1, Q_{t+1} = \underline{1}$


JK Latch						JK LATCH with SET/RESET				
E	J	K	Q_{t+1}	$\bar{Q}_{t+1}$		SET	CLEAR	Q_{t+1}	$\bar{Q}_{t+1}$	
0	X	X	Q_t	$\bar{Q}_t$	✓	0	0	Q_t	$\bar{Q}_t$	✓
1	0	0	Q_t	$\bar{Q}_t$	✓	0	1	0	1	✓
1	0	1	0	1	✓	1	0	1	0	✓
1	1	0	1	0	✓	1	1	Invalid	Invalid	✓
1	1	1	$\bar{Q}_t$	Q_t	✓					

2. Modify given NOR Based JK Flip Flop circuit to introduce Asynchronous CLEAR and SET inputs fill table above?

